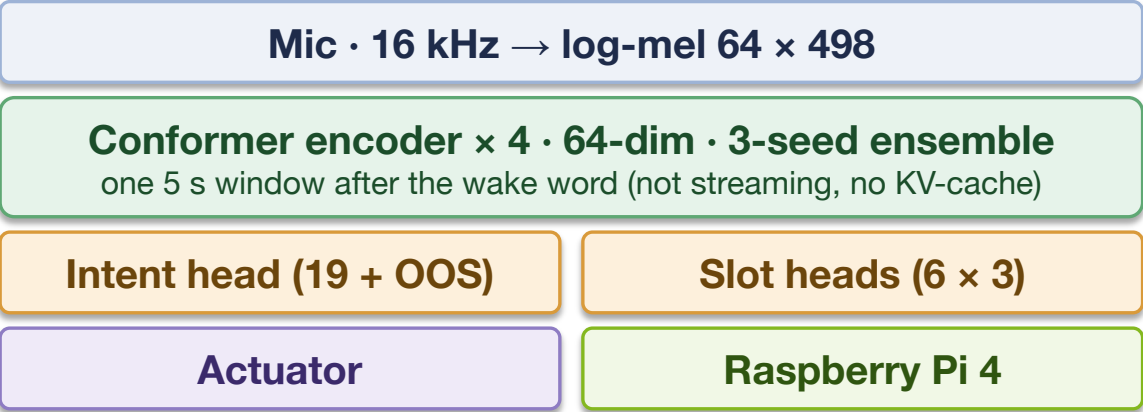


ME2 — Voice Controlled Smart Device

Always-on keyword + intent recognition at the edge · no cloud round-trip

2 October 2026 · [author / team] · Model · Dataset · Training on an RTX 5060 laptop GPU · Validation on RPi 4

Model



Item	Value
Parameters / weights	0.88 M · 4.46 MB fp32 (2.78 MB int8)

Validation on the Raspberry Pi 4

Item	Value
Keyword / intent acc	48.0 % / 46.9 % (97.9 % once woken)
False-accept rate	0.0 % (0 of 10 out-of-scope)
Latency p95 / RTF	1,657 ms e2e (model 101 ms) / 0.020
Runtime	onnxruntime · 2 thr

First live holdout run, laptop speaker → Pi mic (196 + 10 trials). Every miss was a missed wake word; wake model v2 is being retrained and re-run.

Dataset

Item	Value
Source	HF airimonda/ai231-me2-voice-commands
Hours / utts	8.74 h / 15,378 (+1,250 synth. negatives)
Speakers	428 IDs, disjoint train / test / holdout
Labels	19 intents (+ OOS) · 6 slots × 3 values

Group recordings + Common Voice, SLURP, FSC, SNIPS, Speech Commands, TimersAndSuch; 78 % of train is synthetic speech.

Training on the A100 cluster

Item	Value
Cluster	no A100: 1× RTX 5060 Laptop GPU (8 GB)
Objective	CE (intent + masked slot) + misfire term
Optimiser	AdamW · lr 3e-3 · cosine · batch 64
Steps / loss	7,905 / 1.66 (seed 0, best epoch 51)

To Be Submitted

GitHub repository	github.com/airimonda/vcm-me2 – public, MIT
Dataset location	huggingface.co/datasets/airimonda/ai231-me2-voice-commands (public, no DOI yet)
A100 cluster	none: 1× RTX 5060 Laptop GPU · ~12–13 min per seed · seeds 0, 1, 2
Model weights	tag v1.0 in the repo (models/*.onnx, models/ckpt/*.pt) – MIT

What the reviewer will look for

#	Item	Status
1	Repo public, one-command reproduction	Partial: public; commands in docs/paper.md §10
2	Dataset licensed and citable (DOI)	To do: no licence or DOI yet
3	Training logs + final checkpoint committed	Done: results/runs/, models/ckpt/
4	Pi 4 latency reproduced by the posted script	Done: scripts/bench_onnx.py
5	Held-out test set, unseen speakers	Done: 0 speakers shared
6	Baseline of comparable size compared	Done: DS-CNN M 0.861 vs 0.935